do280-4.18: Days remaining **89** | Lab hours used **4/80**

DO280-4.22: Days remaining **89** | Lab hours used **0/80**

Red Hat OpenShift Administration II: Configuring a Production Cluster

Course | Lab Environment

Lab: Deploy Packaged Applications

Deploy and update applications from resource manifests that are packaged for sharing and distribution.

Outcomes

- Deploy an application and its dependencies from resource manifests that are packaged as a Helm chart.
- Update the application to a later version by using the Helm chart.
- Use a container image in a private container registry instead of a public registry.
- Customize the deployment to add resource requests and limits.

As the `student` user on the `workstation` machine, use the `lab` command to prepare yo environment for this exercise, and to ensure that all required resources are available:

```
[student@workstation ~]$ lab start packaged-review
```


Instructions

The API URL of your OpenShift cluster is `https://api.lab.example.com:6443`, and the `oc` command is already installed on your `workstation` machine. You can log in to the OpenShift cluster as the `developer` user with `developer` as the password.

1. Log in to the OpenShift cluster and create the `packaged-review` and `packaged-review-prod` projects.

   1.1. Use the `oc` command to log in to the OpenShift cluster as the `developer` user with `developer` as the password.

   ```
   [student@workstation ~]$ oc login -u developer -p developer \
   https://api.lab.example.com:6443
   Login successful.
   ...output omitted...
   ```

   1.2. Create the `packaged-review` project.

   ```
   [student@workstation ~]$ oc new-project packaged-review
   Now using project "packaged-review" on server ...
   ...output omitted...
   ```

   1.3. Create the `packaged-review-prod` project.

   ```
   [student@workstation ~]$ oc new-project packaged-review-prod
   Now using project "packaged-review-prod" on server ...
   ...output omitted...
   ```

   Hide Solution

2. Add the classroom Helm repository at the `http://helm.lab.example.com/charts` URL and examine its contents. Use `do280-repo` for the name of the repository.

   2.1. Use the `helm repo list` command to list the repositories that are configured for the `student` user.

   ```
   [student@workstation ~]$ helm repo list
   NAME        URL
   do280-repo  http://helm.lab.example.com/charts
   ```

   If the `do280-repo` repository is present, then continue to the next step. Otherwise, add the repository.

   ```
   [student@workstation ~]$ helm repo add \
   do280-repo http://helm.lab.example.com/charts
   "do280-repo" has been added to your repositories
   ```

2.2. Use the `helm search` command to list all the chart versions in the repository.

The `etherpad` chart has versions 0.0.6 and 0.0.7. This chart is a copy of a chart from the `https://github.com/redhat-cop/helm-charts` (`https://github.com/redhat-cop/helm-charts`) repository.

```
[student@workstation ~]$ helm search repo --versions
NAME                    CHART VERSION  APP VERSION  DESCRIPTION
do280-repo/etherpad     0.0.7          latest       ...
do280-repo/etherpad     0.0.6          latest       ...
...output omitted...
```

Hide Solution

3. Install the 0.0.6 version of the `etherpad` chart in the `packaged-review` namespace, with the `test` release name. Use the `registry.lab.example.com:8443/etherpad/etherpad:1.8.17` image in the offline classroom registry.

Create a `values-test.yaml` file with the image repository, name, and tag.

Field	Value
`image.repository`	`registry.lab.example.com:8443/etherpad`
`image.name`	`etherpad`
`image.tag`	`1.8.17`

3.1. Switch to the `packaged-review` project.

```
[student@workstation ~]$ oc project packaged-review
Now using project "packaged-review" on server ...
```

3.2. Examine the values of the chart.

You can configure the image, the deployment resources, and other values. By default, the chart creates a route.

```
[student@workstation ~]$ helm show values do280-repo/etherpad --versi
on 0.0.6
# Default values for etherpad.
replicaCount: 1

defaultTitle: "Labs Etherpad"
defaultText: "Assign yourself a user and share your ideas!"

image:
  repository: etherpad  (1)
  name:  (2)
  tag:  (3)
  pullPolicy: IfNotPresent

...output omitted...

route:
  enabled: true
  host: null  (4)
  targetPort: http

...output omitted...

resources: {}  (5)
...output omitted...
```

1. The registry with the container image.
2. Container image name.
3. Container image tag.
4. Hostname for the OpenShift route resource.
5. The resource requests and limits for this workload. By default, the chart sets this value to `{}`, which indicates an empty map.

3.3. With the default configuration, the chart uses the `docker.io/etherpad/etherpad:latest` container image.

This image is not suitable for the classroom environment. Use the `registry.lab.example.com:8443/etherpad/etherpad:1.8.17` container image instead.

Create a `values-test.yaml` file with the following content:

```
image:
  repository: registry.lab.example.com:8443/etherpad
  name: etherpad
  tag: 1.8.17
```


3.4. Install the `etherpad` chart in the `packaged-review` namespace.

- Use the `values-test.yaml` file that you created in the previous step.
- Use `test` as the release name.

```
[student@workstation ~]$ helm install test do280-repo/etherpad \
-f values-test.yaml --version 0.0.6
NAME: test
LAST DEPLOYED: Mon Jul 27 03:00:30 2026
NAMESPACE: packaged-review
STATUS: deployed
REVISION: 1
TEST SUITE: None
```

3.5. Use the `helm list` command to verify the installed version of the `etherpad` chart.

```
[student@workstation ~]$ helm list
NAME  NAMESPACE        REVISION  ...  STATUS    CHART           APP V
ERSION
test  packaged-review  1         ...  deployed  etherpad-0.0.6  lates
t
```

3.6. Verify that the pod is running and that the deployment is ready.

```
[student@workstation ~]$ watch oc get deployments,pods
Every 2.0s: oc get deployments,pods                    ...

NAME                             READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/test-etherpad    1/1     1            1           27s

NAME                                 READY   STATUS    RESTARTS   AGE
pod/test-etherpad-b3356d443-5dfa9    1/1     Running   0          27s
```

Wait until the pod is ready. Press **Ctrl**+**C** to exit the `watch` command.

3.7. Verify that the pod executes the specified container image.

```
[student@workstation ~]$ oc describe pods -n packaged-review | \
grep -E '^Name:|Image:'
Name:             test-etherpad-b3356d443-5dfa9
    Image:          registry.lab.example.com:8443/etherpad/etherpad:
1.8.17
```

3.8. Get the route to obtain the application URL.

```
[student@workstation ~]$ oc get routes
NAME           HOST/PORT
...
test-etherpad  test-etherpad-packaged-review.apps.lab.example.com
...
```

3.9. Open a web browser and go to the following URL to view the application page.

`https://test-etherpad-packaged-review.apps.lab.example.com`

Hide Solution

4. Upgrade the `etherpad` application in the `packaged-review` namespace to the 0.0.7 version of the chart. Set the image tag for the deployment in the `values-test.yaml` file.

Field	Value
`image.tag`	`1.8.18`

4.1. Edit the `values-test.yaml` file and update the image tag value.

```
image:
  repository: registry.lab.example.com:8443/etherpad
  name: etherpad
  tag: 1.8.18
```

4.2. Use the `helm search` command to verify that the repository contains a more recent version of the `etherpad` chart.

```
[student@workstation ~]$ helm search repo --versions etherpad
NAME                  CHART VERSION  APP VERSION  DESCRIPTION
do280-repo/etherpad   0.0.7          latest       ...
do280-repo/etherpad   0.0.6          latest       ...
```

4.3. Use the `helm upgrade` command to upgrade to the latest version of the chart.

```
[student@workstation ~]$ helm upgrade test do280-repo/etherpad \
-f values-test.yaml --version 0.0.7
Release "test" has been upgraded. Happy Helming!
NAME: test
LAST DEPLOYED: Mon Jul 27 03:01:09 2026
NAMESPACE: packaged-review
STATUS: deployed
REVISION: 2
TEST SUITE: None
```

4.4. Use the `helm list` command to verify the installed version of the `etherpad` chart.

```
[student@workstation ~]$ helm list
NAME  NAMESPACE        REVISION  ...  STATUS    CHART           APP V
ERSION
test  packaged-review  2         ...  deployed  etherpad-0.0.7  lates
t
```

4.5. Verify that the pod is running and that the deployment is ready.

```
[student@workstation ~]$ watch oc get deployments,pods
Every 2.0s: oc get deployments,pods                    ...

NAME                             READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/test-etherpad    1/1     1            1           3m31
s

NAME                                  READY   STATUS    RESTARTS   AGE
pod/test-etherpad-43s558ah8f-bhtur    1/1     Running   0          64s
```

Wait until the pod is ready. Press **Ctrl+C** to exit the `watch` command.

4.6. Verify that the pod executes the updated container image.

```
[student@workstation ~]$ oc describe pods -n packaged-review | \
grep -E '^Name:|Image:'
Name:             test-etherpad-43s558ah8f-bhtur
    Image:          registry.lab.example.com:8443/etherpad/etherpad:
1.8.18
```

4.7. Reload the `test-etherpad` application welcome page in the web browser.

Hide Solution

5. By using version `0.0.6`, create a second deployment of the chart in the `packaged-review-prod` namespace, with the `prod` release name. Copy the `values-test.yaml` file to the `values-prod.yaml` file, and set the route host.

Field	Value
`route.host`	`etherpad.apps.lab.example.com`

Access the application in the route URL to verify that it is working correctly:

`https://etherpad.apps.lab.example.com`

5.1. Switch to the `packaged-review-prod` project.

```
[student@workstation ~]$ oc project packaged-review-prod
Now using project "packaged-review-prod" on server ...
```

5.2. Copy the `values-test.yaml` file to the `values-prod.yaml` file.

```
[student@workstation ~]$ cp values-test.yaml values-prod.yaml
```

5.3. Set the route host in the `values-prod.yaml` file.

```
image:
  repository: registry.lab.example.com:8443/etherpad
  name: etherpad
  tag: 1.8.18
route:
  host: etherpad.apps.lab.example.com
```

5.4. Install the 0.0.6 version of the `etherpad` chart in the `packaged-review-prod` namespace.

Use the `values-prod.yaml` file that you edited in the previous step. Use `prod` as the release name.

```
[student@workstation ~]$ helm install prod do280-repo/etherpad \
-f values-prod.yaml --version 0.0.6
NAME: prod
LAST DEPLOYED: Mon Jul 27 03:01:09 2026
NAMESPACE: packaged-review-prod
STATUS: deployed
REVISION: 1
TEST SUITE: None
```

5.5. Use the `helm list` command to verify the installed version of the `etherpad` chart.

```
[student@workstation ~]$ helm list
NAME  NAMESPACE            REVISION  ...  STATUS    CHART
APP VERSION
prod  packaged-review-prod  1         ...  deployed  etherpad-0.0.6
latest
```

5.6. Verify that the pod is running and that the deployment is ready.

```
[student@workstation ~]$ watch oc get deployments,pods
Every 2.0s: oc get deployments,pods                    ...

NAME                              READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/prod-etherpad     1/1     1            1           65s

NAME                                    READY   STATUS    RESTARTS   AGE
pod/prod-etherpad-744f5b4bff-7tx45      1/1     Running   0          65s
```

Wait until the pod is ready. Press **Ctrl**+**C** to exit the `watch` command.

5.7. Verify that the pod executes the specified container image.

```
[student@workstation ~]$ oc describe pods -n packaged-review-prod | \
grep -E '^Name:|Image:'
Name:             prod-etherpad-744f5b4bff-7tx45
    Image:          registry.lab.example.com:8443/etherpad/etherpad:
1.8.18
```

5.8. Verify the deployment by opening a web browser and going to the application URL. This URL corresponds to the host that you specified in the `values-prod.yaml` file. The application welcome page appears in the production URL.

`https://etherpad.apps.lab.example.com`

Hide Solution

6. Add limits to the `etherpad` instance in the `packaged-review-prod` namespace. The chart values example contains comments that show the required format for this change. Set limits and requests for the deployment in the `values-prod.yaml` file. Use version `0.0.7` of the chart.

Field	Value
`resources.limits.memory`	`256Mi`
`resources.requests.memory`	`128Mi`

6.1. Edit the `values-prod.yaml` file. Configure the deployment to request 128 MiB of RAM, and limit RAM usage to 256 MiB.

```
image:
  repository: registry.lab.example.com:8443/etherpad
  name: etherpad
  tag: 1.8.18
route:
  host: etherpad.apps.lab.example.com
resources:
  limits:
    memory: 256Mi
  requests:
    memory: 128Mi
```

6.2. Use the `helm upgrade` command to upgrade to the latest version of the chart.

```
[student@workstation ~]$ helm upgrade prod do280-repo/etherpad \
-f values-prod.yaml --version 0.0.7
Release "prod" has been upgraded. Happy Helming!
NAME: prod
LAST DEPLOYED: Mon Jul 27 03:09:07 2026
NAMESPACE: packaged-review-prod
STATUS: deployed
REVISION: 2
TEST SUITE: None
```

6.3. Verify that the pod is running and that the deployment is ready.

```
[student@workstation ~]$ watch oc get deployments,pods
Every 2.0s: oc get deployments,pods                    ...

NAME                            READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/prod-etherpad   1/1     1            1           3m14
s

NAME                                  READY   STATUS    RESTARTS   AGE
pod/prod-etherpad-5ab75tccgd-b7mnp    1/1     Running   0          36s
```

Wait until the pod is ready. Press **Ctrl+C** to exit the `watch` command.

6.4. Examine the application pod from the production instance of the application to verify the configuration change.

```
[student@workstation ~]$ oc describe pods -n packaged-review-prod | \
grep -E -A1 '^Name:|Limits|Requests'
Name:             prod-etherpad-5ab75tccgd-b7mnp
Namespace:        packaged-review-prod
--
    Limits:
      memory:  256Mi
    Requests:
      memory:   128Mi      14/72
```

6.5. Examine the pod of the test instance of the application in the `packaged-review` namespace. This deployment uses the values from the `values-test.yaml` file, which does not specify resource limits or requests. The pod in the `packaged-review` namespace does not have a custom resource allocation.

```
[student@workstation ~]$ oc describe pods -n packaged-review | \
grep -E -A1 '^Name:|Limits|Requests'
Name:             test-etherpad-59d775b78f-ftmsz
Namespace:        packaged-review
```

6.6. Use the `helm list` command to verify the installed version of the `etherpad` chart.

```
[student@workstation ~]$ helm list
NAME  NAMESPACE             REVISION  ...  STATUS    CHART
APP VERSION
prod  packaged-review-prod  2         ...  deployed  etherpad-0.0.7
latest
```

6.7. Reload the application welcome page in the web browser. The deployment continues working after you add the limits.

Hide Solution

Evaluation

As the `student` user on the `workstation` machine, use the `lab` command to grade your work. Correct any reported failures and rerun the command until successful.

```
[student@workstation ~]$ lab grade packaged-review
```

Finish

As the `student` user on the `workstation` machine, use the `lab` command to complete this exercise. This step is important to ensure that resources from previous exercises do not impact upcoming exercises.

```
[student@workstation ~]$ lab finish packaged-review
```

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